

Determining the Verdet constant of SF57 glass using phase-sensitive detection

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Abstract

We set out to determine the Verdet constant of SF57 glass and develop the techniques to reduce the uncertainties in the measurement, in the hope that this will lead to more compact and accurate magneto-optical devices such as isolators. We measure the angle of deflection that polarised light emitted by a LED experiences passing through glass in a magnetic field of a known magnitude. Due to a small bending angle, we attempt to use Malus' law and measure the change in intensity at the analyser to deduce the deflection angle. We modulate the magnetic field across the glass to remove systematic errors and to separate the measurement from low frequency noise. We use phase-sensitive detection to optimise the alignment of the polariser and analyser for the precision of our measurement. By recording the fluctuations in voltage across the photodiode, we calculate the Verdet constant of SF57 as $1.93 \times 10^{-4} \pm 4 \times 10^{-6} \text{ rad A}^{-1}$ which is inconsistent with the value presented by previous literature of $9.04 \times 10^{-5} \text{ rad A}^{-1}$. The uncertainty of the value obtained by using phase sensitive detection to align the analyser is 4% less than with previous measurements made without this technique. We hope that this method can be used to gain knowledge of the properties of magneto-optical glass which will encourage its future use in optical devices.

1 Introduction

Materials that have magneto-optical properties like the Faraday effect are important to the fields of optical engineering and glass science. Crystals with these properties have been studied but research into glasses has been limited. Glasses are transparent in the visible and near infrared regions. They can be made into complex shapes unlike many crystals [1]. Developing glasses that exhibit the Faraday effect could increase the efficiency of isolators and rotators because less light is absorbed, and devices can be shaped to match magnetic field distributions. The ability to shape glass through moulding and drawing processes supports systems that require magneto-optical media of specific geometries.

Isolators are devices that prevent the back reflection of light in a system. They are crucial to lasers used in fibre optical cables, industrial cutting and quantum optics. Inefficiencies in an isolator can cause heating up which alters the deflection angle of light, leading to the failure of the laser. Rotators are used in magnetic field sensors in MRI scanners and particle accelerators. Inefficiencies can give inaccurate readings of

weak fields crucial to the working of these machines. For such reasons, it is productive to investigate the use of glasses in such devices, namely through the determination of the Verdet constant of a glass: a measure of the deflection of light for a given magnetic field. Attempts have been made to find this constant but current techniques give rise to uncertainties that limit the compactness of devices using magneto-optical glasses.

Light is deflected in glass by the order of a degree or less. It is passed through a polarimeter and the change in intensity at the analyser is measured. However, the change is very small and efforts to measure it lacked in enough precision to be of any use. It turns out that modulating the magnetic field along the glass leads to fluctuations in the voltage measured across the photodiode receiving the light. These variations are proportional to the deflection of the light beam and are maximised when the analyser is aligned 45 degrees from the polariser. Recording the fluctuations yield a Verdet constant for glass to a much lower uncertainty. The aim of our investigation is to build upon this effort and use phase sensitive detection to more accurately align the polarimeter and reduce the uncertainty in the Verdet constant of SF57 glass. Any reduction in the uncertainty of this value will lead to more compact magneto-optical glass media which can be used in a greater range of optical devices at a lower cost. The possible improvement of performance in instruments across many fields renders our research pivotal.

2 Theoretical background

In 1845, Michael Faraday found that shining a beam of light into a rod with a magnetic field of strength, H , along it, parallel to the propagation direction, caused a deflection of the beam, α . If the glass has length d , and a Verdet constant ϑ , it can be shown empirically that:

$$\alpha = H\vartheta d$$

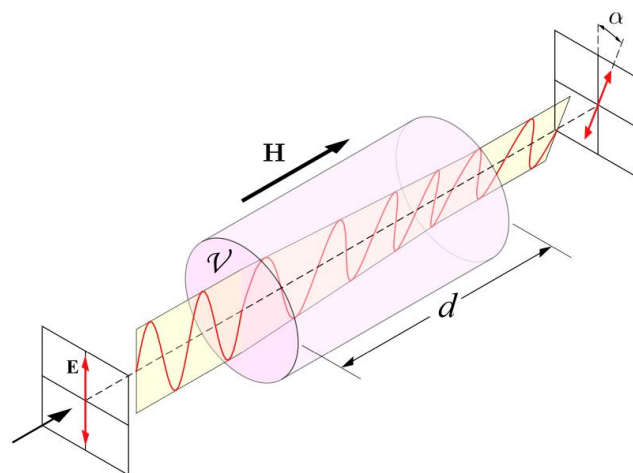


Fig 1: Deflection, α , of a light beam passing through glass rod of distance d in magnetic field H [2]

Since it is difficult to measure the deflection angle directly, we use Malus' law for polarised light to use the change in intensity of light at the analyser to infer the deflection angle.

$$F(\alpha) = F_0 \cos^2(\theta + \alpha)$$

F_0 is the incident flux on the analyser and θ is the angle between the polariser and the analyser. With α much less than 1, we use the Taylor expansion to get:

$$F(\alpha) = F_0 (\cos^2(\theta) - \alpha \sin(2\theta) + O(\alpha^2))$$

We apply a sinusoidal magnetic wave, $H = H_0 \cos(wt)$, across the glass to obtain:

$$F(t) = F_0 (\cos^2(\theta) - H_0 \vartheta d \cos(wt) \sin(2\theta) + O(\alpha^2))$$

The intensity on photodiode has a varying component,

$$F_{AC} = F_0 H_0 \vartheta d \cos(wt) \sin(2\theta)$$

and an average component:

$$F_{DC} = F_0 \cos^2(\theta)$$

We see that the AC component is maximised when $\theta = (2n + 1)\frac{\pi}{4}$ radians. We don't have F_0 but we can use the ratio of the 2 components to find the Verdet constant:

$$\frac{F_{AC}}{F_{DC}} = 2\vartheta d H_0 \tan(\theta)$$

The glass is kept in a solenoid, so the magnetic field strength across it has the form:

$$H = \frac{NI}{L}$$

N is the number of coils in the solenoid, I is the current flowing through it and L is the length of the solenoid. In our experiment, we will find the exact cross polarisation position of the polarimeter by dithering with phase sensitive detection. This ensures that we can align the analyser very close to 45 degrees away from the polariser when measuring the intensity on the photodiode. If we include higher order terms to the expression of our intensity, we see:

$$F(\alpha) = F_0 (\cos^2(\theta) - \alpha \sin(2\theta) - \alpha^2 \cos(2\theta) + O(\alpha^3))$$

If we plug in $\alpha = a \cos(wt)$, with $a = H_0 \vartheta d$, then:

$$F(\alpha) = F_0 \left(\cos^2(\theta) - a \cos(\omega t) \sin(2\theta) - a^2 \frac{(\cos(2\omega t) + 1)}{2} \cos(2\theta) + O(\alpha^3) \right)$$

If the analyser is exactly cross polarised with $\theta = 90^\circ$, then the first 2 terms of expression disappear and we have a 2ω intensity signal. If 2θ dips below 180° , then the second term is negative and third term is positive. The $\cos(\omega t)$ term is antiphase with the driving magnetic force. a is very small so $a^2 \ll a$ and so $\cos(\omega t)$ dominates as in Fig 2 c). If 2θ dips above 180° as in Fig 2 d), then the second term is positive and third term is positive. The $\cos(\omega t)$ term is in phase with the driving magnetic force. In phase sensitive detection, we multiply the intensity signal on the photodiode with the signal applied across the solenoid to make a *demodulated* signal. If signal 1 is of the form $V_1 \cos(\omega t + \phi_1)$ and signal 2, $V_2 \cos(\omega t + \phi_2)$, then the demodulated signal is:

$$P_{12}(t) = \frac{V_1 V_2 (\cos(\phi_1 - \phi_2) + \cos(2\omega t + \phi_1 + \phi_2))}{2}$$

Over many integer cycles, the component of frequency 2ω goes to zero and we are left with:

$$\langle P_{12} \rangle = \frac{V_1 V_2 \cos(\phi_1 - \phi_2)}{2}$$

Considering the intensity signal across the photodiode and the applied magnetic field, the resulting demodulated signal will be zero when the polarimeter is exactly cross polarised and will switch signs on either side of cross-polarisation. Using this information, we can determine the exact cross polarisation position to a high degree of precision.

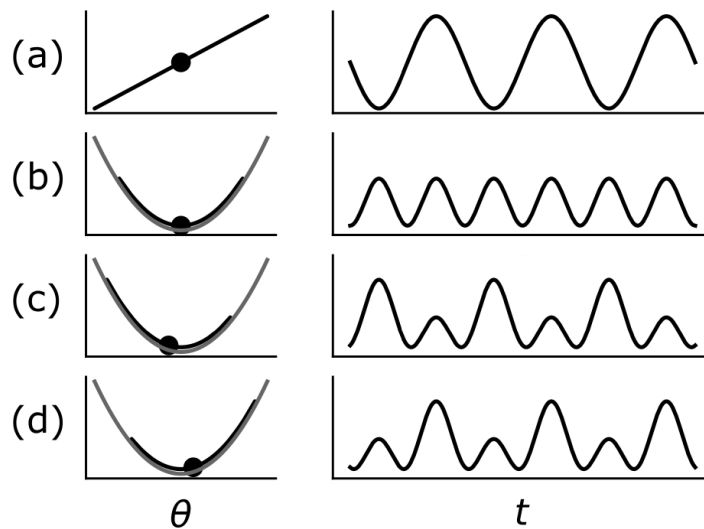


Fig 2: Plots of intensity against 1) polariser-analyser angle on the left and 2) time on the right [2]

3 Method

3.1 Experimental design

An LED shines light onto a collimator lens. The light hits a polariser and then travels through the SF57 glass rod placed in a solenoid. The deflected light is incident on an analyser and then is focused onto a photodiode by a convex lens. We use one PCB that is connected to a power supply and a signal generator to drive the current through the solenoid. Another PCB, also connected to a power supply drives a current through the LED and receives the signal from the photodiode. A crocodile clip connects the test point with this signal to a picoscope, which displayed an alternating voltage signal on top of a DC voltage. We measured the ratio of the 0 to peak AC voltage variation against the DC voltage level which is the same as $\frac{F_{AC}}{F_{DC}}$. Firstly, we set analyser position to $\theta = 45^\circ$ and estimated V_{AC} by using the voltage scale on scope mode. Then we made the same estimate using spectral mode. We precisely determined the cross-polarisation position of the polarimeter and finally obtained a precise Verdet constant for the glass from scope mode.



Fig 3: Experimental apparatus. From left to right: LED, lens, polariser, glass in solenoid, analyser, lens, photodiode [2]

3.2 Equipment

The solenoid has 2200 turns. SF57 is a flint glass with a relatively high Verdet constant, a diameter of 23 mm and a length of 50 mm. The LED has a peak wavelength of $405 \pm 2.5 \text{ nm}$. The light signal from the photodiode is a weak current, so we make a transimpedance amplifier of gain 470000 to convert the current into a larger voltage as shown in Fig 4.

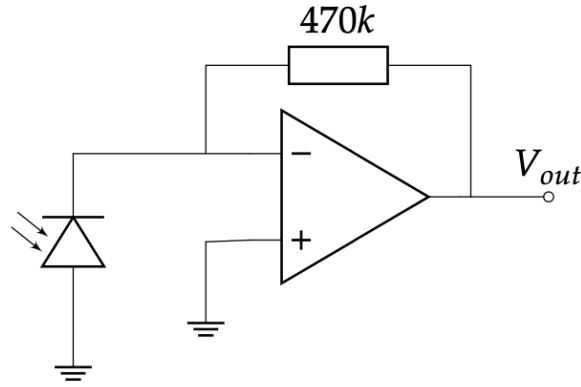


Fig 4: Schematic diagram of photodiode transimpedance amplifier [2]

3.3 Procedure

At the beginning of our experiment, we aligned our photodiode with the LED to observe an image of the semiconductor inside the LED on the diode and checked if it was on the centre of the diode. We rotated the analyser to produce the minimum intensity of light hitting the 2nd lens by observing the light leaving the analyser. This occurred when the analyser was approximately at an angle 45° and 225° to the vertical, indicating that the polariser was aligned at 45° to the vertical. To determine the magnetic field strength caused by the solenoid in the glass, we measured the voltage across a $1\ \Omega$ resistor in series with the coil by displaying the voltage across it on the picoscope. We used $H = \frac{NI}{L}$ to find H. The voltage amplitude from signal generator was adjusted until the coil current was varying with a maximum peak to peak amplitude of 17mA without any clipping. The frequency of current across coil was set to 70Hz. Our apparatus was covered with a black shelter to block background light from hitting the photodiode. We then adjusted the analyser position to make $\theta = 45^\circ$ for maximum AC voltage variation on photodiode and make measurements of V_{AC} and V_{DC} from scope mode. Then, we investigated the relationship between the magnitude of the current driving the coil and the peak to peak AC voltage across the photodiode. We confirmed that the relationship was linear, but the error in V_{AC} grew for low driving currents due to the position of peaks in signal varying with time. This motivates the measurement of V_{AC} from spectral mode.

In spectral mode, we read the magnitude of peak at 70Hz to find V_{AC} because we knew the frequency of AC signal across diode had the same frequency as the current across the coil. V_{DC} was determined from magnitude of voltage of peak at 0 Hz on spectrum. T

he window function in spectral mode was set to 'flat top' and we used a high maximum frequency of 100kHz so that sampling rate was high enough to avoid the aliasing of high frequency noise to our AC signal. The spectral mode was a time average, and we waited for about 30 seconds before reading the magnitudes of frequency peaks.

To gain a more precise measurement of the Verdet constant of the glass, we had to determine the exact cross polarisation position of the polarimeter. We used a python program to run phase sensitive detection. It would output the average of the demodulated signal of V_{AC} from the photodiode and the signal driving the coil. By starting off with the analyser close to the cross-polarisation position, we varied the analyser position in small increments, trying to reduce the amplitude of the average of the demodulated signal as much as possible, reading the angle with the vernier scale on the analyser mount. The DC average of demodulated signal was read 30 seconds after we started the program. After this, we investigated the effect of crosstalk between the 2 PCBs driving the coil current and receiving the photodiode signal by looking for peaks at 70Hz on the spectrum of voltage signal received from photodiode when one PCB performed both functions and the LED output was zero.

Using both spectral and scope modes to find V_{AC} , we use the following equation to determine the Verdet constant:

$$\frac{V_{AC}}{V_{DC}} = 2\vartheta d H_0 \tan(\theta)$$

4 Results

4.1 Estimating the Verdet constant

First, we measured the peak voltage across the resistor in series with the coil to be $110 \pm 1 \text{ mV}$, implying that the peak current, I_0 , into the coil was $0.11 \pm 0.001 \text{ A}$. Using scope mode, we saw that $V_{AC} = 19.5 \pm 0.2 \text{ mV}$ and $V_{DC} = 486.7 \pm 0.5 \text{ mV}$. With $d = 50 \pm 0.3 \text{ mm}$, $\theta = 45 \pm 0.5^\circ$ and the number of coils, $N = 2200 \pm 1$, we combine $H_0 = \frac{NI_0}{L}$ and equation 7 to calculate $\vartheta = 1.6 \times 10^{-4} \pm 1.1 \times 10^{-5} \text{ rad A}^{-1}$.

We can make the same calculation again with spectral mode and obtain a more precise Verdet constant. The magnitude of the peak corresponding to V_{AC} is $-79.19 \pm 0.5 \text{ dB}$, giving $V_{AC} = 7.8 \times 10^{-5} \pm 6 \times 10^{-6} \text{ V}$. $V_{DC} = 0.0964 \pm 0.001 \text{ V}$ and using the method above, $I_0 = 0.105 \pm 0.001 \text{ A}$. The same analysis yields $\vartheta = 4.9 \times 10^{-5} \pm 3 \times 10^{-6} \text{ rad A}^{-1}$.

4.2 Determining the precise Verdet constant

Using phase sensitive detection, we obtained the series of values shown in table 1 and plotted in figure 5. With a line of best fit on our graph to find the x-intercept, we found the cross-polarisation angle to be $\theta = 49.32 \pm 0.08^\circ$. Using scope mode, we recorded $V_{AC} = 21.7 \pm 0.3 \text{ mV}$, $V_{DC} = 173.1 \pm 0.5 \text{ mV}$ and $I_0 = 0.28 \pm 0.005 \text{ A}$. Using the analysis above, we obtain $\vartheta = 1.93 \times 10^{-4} \pm 4 \times 10^{-6} \text{ rad A}^{-1}$.

Angle / °	DC Average of demodulated signal/ V
48.83	$1.1 \times 10^{-4} \pm 2 \times 10^{-5}$
49.08	$5.44 \times 10^{-5} \pm 3 \times 10^{-5}$
49.25	$2.5 \times 10^{-5} \pm 2 \times 10^{-5}$
49.33	$-9.9 \times 10^{-6} \pm 1.2 \times 10^{-5}$
49.42	$-1.77 \times 10^{-5} \pm 3.1 \times 10^{-5}$
50.17	$-9.9 \times 10^{-5} \pm 2.1 \times 10^{-5}$

Table 1: Polariser-Analyser angle and the DC average of the resulting demodulated signal

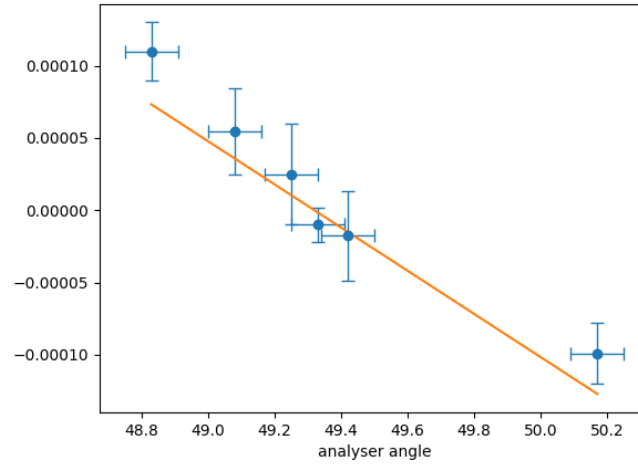


Fig. 5: DC average of demodulated signal against cross polarisation angle

5 Discussion

The aim of this investigation was to develop techniques that could measure the Verdet constant of SF57 glass to a higher precision. Using a modulated magnetic field goes a long way to removing noise from our measurement. The voltage signal across the photodiode experiences a lot of low frequency noise from the current received from the photodiode and that driving the LED. There is the random trapping and releasing of electrons due to defects in the wire. Defects themselves can move causing mobility fluctuations of electrons. The environment introduces mechanical vibrations from footsteps, acoustic noise etc. as well as thermal drift and thermal fluctuations in components. If the magnetic field across the glass was constant, we would be reading a peak at 0 Hz on the frequency spectrum corresponding to a single intensity change from the deflection in the glass. This reading would be plagued by all the noise mentioned above. By modulating the magnetic field across the glass, we move our measurement into a higher frequency region that is less affected by noise. In addition to this, the differential measurement we make will remove any systematic errors that arise from our setup.

Placing the apparatus under a black cover removed the DC offset from background light in the room. However, with the LED brightness set to zero, there was also a small DC offset from the transimpedance amplifier used with the photodiode which we did not remove from the DC voltage signal but could have been to improve our results. In spectral mode, high frequency noise peaks were seen at 115kHz and 240kHz due to resonant oscillations in the circuits as well as at 60kHz and 80 kHz due to background light. However, because the frequency of V_{AC} was much lower than this, we don't suspect this noise affected our results in anyway. A limitation of the setup was that the current passing through the solenoid heated it up, increasing its resistance, affecting the current in the coil and the magnetic field it produces. An improvement would've been to turn off the current in the coil between measurements and perhaps even to keep the setup refrigerated to maintain a constant temperature. When checking for crosstalk between the signals when only one PCB was used, we did find a peak at the frequency of the coil current in the spectrum of the signal from the photodiode even when the LED brightness was set to zero. This confirms our decision to use 2 different PCBs to drive the coil current and to receive the photodiode signal.

We also see that the Verdet constant obtained using spectral mode on the oscilloscope is more precise than using scope mode. There are several reasons for this. In spectral mode, we average the signal over many cycles to find magnitude of V_{AC} whereas in scope mode, we use a single cycle. Spectral mode lets us read V_{AC} from a single measurement rather than recording 2 points at a crest and a trough. The signal to noise ratio is much less in spectral mode, where the V_{AC} peak is separated from low frequency noise, making it easier to record. In both estimations above, the polariser-analyser was set by finding the cross polarised position by eye, checking when the light transmitted through the analyser was at a minimum. A major source of uncertainty in this method is in θ , being known to $\pm 0.5^\circ$. The relevant quantity, $\tan(\theta)$, has an uncertainty of ± 0.017 . Using the dithering procedure outlined in section 3.3, we found the cross-polarisation angle to an error of $\pm 0.08^\circ$, giving $\tan(\theta) \pm 0.003$ which has an absolute uncertainty of about 6 times less than from our estimate. This reduces the error of the Verdet constant of SF57 glass substantially from 6% to about 2%. Previous literature shows the Verdet constant of SF57 glass to be $9.04 \times 10^{-5} \text{ rad A}^{-1}$ which is not consistent with our measurement. This may be due to the limitations of the experiment outlined in this section. Regardless, the use of phase sensitive detection increased the precision of our value by a large margin, and we hope this technique can be used for other experiments that look at the magneto-optical properties of glass.

6 Conclusion

In this experiment we found the Verdet constant of SF57 glass with light of $\lambda = 405 \pm 2.5 \text{ nm}$ to be $\vartheta = 1.93 \times 10^{-4} \pm 4 \times 10^{-6} \text{ rad A}^{-1}$ which is inconsistent with the value from previous literature of $\vartheta = 9.04 \times 10^{-5} \text{ rad A}^{-1}$. The substantial improvement in precision of our value for the Verdet constant showcases the potency of phase sensitive detection in measuring small Faraday deflections. The inconsistency of our value with that previously accepted may be due to inaccuracies caused by not accounting for the DC offset caused by our transimpedance amplifier as well as any temperature drift caused by heating up of the solenoid. Temperature control and further measures to remove systematic errors would improve the accuracy of this experiment.

References

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2. 'Michaelmas Term Class Manual: Part II', 2025, Laboratory manual: Department of Physics, University of Cambridge.